| Method | Memory | Budget | Mean scan time | MAPE % [95% CI] | Scans < 5% | Worst scan MAPE % |
|---|---|---|---|---|---|---|
| **RL** | **TI histogram** | **560 s** | **541 s** | **2.93 [2.58, 3.30]** | **95.8%** | **5.13** |
| RL | none | 560 s | 548 s | 4.16 [3.41, 4.99] | 66.7% | 9.50 |
| RL | LSTM[†] | 560 s | 525 s | 7.31 [6.10, 8.65] | 25.0% | 15.02 |
| Fixed, distilled from RL policy[§] | – | 560 s | 549 s | 3.34 [2.30, 4.86] | 91.7% | 17.87 |
| Fixed log TI grid | – | 560 s | 504 s | 6.04 [5.36, 6.70] | 25.0% | 10.29 |
| Cramér-Rao optimal | – | 560 s | 558 s | 7.74 [7.08, 8.42] | 8.3% | 11.72 |
| RL | none[‡] | 240 s | 227 s | 4.62 [4.06, 5.19] | 62.5% | 7.74 |
| Fixed log TI grid | – | 240 s | 163 s | 6.86 [5.83, 7.90] | 20.8% | 12.04 |
| Cramér-Rao optimal | – | 240 s | 240 s | 15.96 [12.26, 19.81] | 0.0% | 35.53 |

[†] Undertrained: the LSTM ran 4.6× slower, so within the same 9 h budget it took 7.3× fewer full-Bloch steps. Its best checkpoint predates full-Bloch training, so this is not a fair test of recurrent memory.

[‡] Policy also chose the flip angle.

[§] One fixed schedule: per-block median TI and TR of the TI-histogram policy's schedules on 24 validation scans (13 blocks, 549 s).